Deliverable 2: Refined Use Case Descriptions

ParkChoiceSG Lab 2 analysis revision aligned with the current boundary-class handover and four-view entity model.

Revision date: 14 September 2026

Scope item	Current decision
Use-case identifiers	UC-01 through UC-09 remain the assigned traceability identifiers. Device-local FR-1.x, FR-2.4/2.6, FR-6.x and FR-7.x capabilities are represented in the BCE model and dialog map using their SRS functional-requirement references.
Model alignment	Arrival context, origin, ordered ranking criteria, prediction fallback, anonymous quota enforcement, operator authentication and manual refresh follow the current boundary/entity handover.

UC-01 — Request Carpark Recommendation

Field	Specification
Use case ID	UC-01
Use case name	Request Carpark Recommendation
Owner	Nestor
Last updated	14 September 2026
Actors	Anonymous Driver; Parking Data Provider; Destination, Map and Routing Provider
Description	The driver requests ranked carpark recommendations near a destination. The request captures the intended arrival context, origin coordinates, parking duration, search radius and ordered ranking criteria. ParkChoiceSG

	combines verified rates, current or predicted availability and routed travel measures, then explains the weighted ranking.
Preconditions	1. ParkChoiceSG is running and the recommendation interface is available. 2. The selected ranking criteria and vehicle preference are valid, either from the request or device-local preferences.
Postconditions	1. On success, the driver receives one to five explained results with labelled evidence and confidence. 2. On failure, the driver receives a clear reason why a reliable recommendation cannot be produced. 3. No driver account or backend driver profile is created. 4. The installation may retain a local SearchHistoryEntry, capped at ten entries.
Priority	Must Have (core recommendation); prediction and local preferences are supported enhancements
Frequency	High - normally once per parking search or journey
Main flow	1. The driver opens the carpark recommendation function. 2. The driver enters a destination, intended arrival time and parking duration. The application obtains origin coordinates and the active search preferences according to the agreed SRS flow. 3. The system validates duration, arrival context, origin, search radius and the four-value criteriaPriority list, then resolves the destination. 4. The system finds supported carparks within the requested radius and obtains current or historical availability, applicable rate schedules, and walking and driving route measures. 5. The system applies active closure and rate overrides, vehicle restrictions, freshness rules and evidence-eligibility checks. 6. The system estimates cost for the arrival time and duration. It predicts lots when supported, records confidence, and falls back to the current reading when historical support is insufficient. 7. The system computes a weighted score using the ordered Cost, WalkingDistance, DrivingTime and PredictedAvailability criteria, ranks candidates, and returns at most five results. 8. The system displays each result's raw factors, timestamps, confidence or fallback explanation, warnings and concise ranking justification. The driver may select a result to start UC-02.

Alternative flows	**UC-01.AC.1 - Invalid request** 1. The system identifies an invalid duration, arrival time, origin, radius or criteria list. 2. It identifies the field that must be corrected and returns the driver to the search form. **UC-01.AC.2 - Destination cannot be resolved** 1. The destination provider cannot resolve the entered destination unambiguously. 2. The system asks the driver to refine the destination or select a suggested match. 3. The use case resumes after the driver responds. **UC-01.AC.3 - Origin unavailable** 1. The system cannot obtain the required origin coordinates. 2. It asks the driver to grant the agreed location permission or enter/select an origin. If no origin is supplied, the system explains that driving-time ranking is unavailable and follows the SRS fallback rule. **UC-01.AC.4 - Partial provider failure** 1. One provider or part of its data is unavailable. 2. The system continues only with candidates that have sufficient valid evidence for the selected criteria and visibly warns about omitted candidates or factors. **UC-01.AC.5 - Prediction unavailable** 1. There is insufficient comparable history or the prediction service fails. 2. The system uses the current availability reading when valid, marks the fallback and lowers the confidence rather than inventing a prediction. **UC-01.AC.6 - Missing or stale candidate evidence** 1. A candidate lacks current availability, a valid rate, a route measure or another required factor. 2. The system excludes it from a fair comparison and records the reason. **UC-01.AC.7 - Fewer than five suitable carparks** 1. Fewer than five candidates satisfy the eligibility rules.

	2. The system displays every suitable result and states that no additional comparable carparks were found. **UC-01.AC.8 - No suitable carpark** 1. No candidate has enough valid evidence for the selected criteria. 2. The system explains the unavailable factors and suggests changing the search or retrying later.
Exceptions	**UC-01.EX.1 - Required data sources unavailable** 1. The system cannot obtain enough reliable evidence. 2. It informs the driver that recommendations are temporarily unavailable and offers a retry. **UC-01.EX.2 - Unexpected system error** 1. The system records the error without exposing credentials or sensitive provider details. 2. It informs the driver that the request failed and offers a retry.
Includes	None.
Special requirements	● When required dependencies respond within one second, at least 95% of recommendation requests should display within three seconds. ● Availability is labelled as current reported availability when it is a current reading; a provider timestamp is shown when supplied, otherwise the displayed time is explicitly labelled retrieval time. ● Every result exposes the factors used, weighted score, prediction lots and confidence, estimated-travel flag, warnings and ranking explanation. ● Missing or stale evidence is never replaced with a fabricated zero or presented as current. ● The four RankingCriterion values occur once each in criteriaPriority and retain their order. ● The driver is not required to create an account. Device-local preferences, history and other local features remain on the installation.

Assumptions	• The supported arrival-time horizon, source freshness thresholds, score normalisation and tie-break rule are recorded in the active SRS and tests. • The driver is searching for an ordinary passenger car unless a supported VehicleType preference is selected. • Estimated cost is based on verified published rules and is not a payment quotation or guarantee. • Current or predicted availability is evidence, not a reservation or guarantee of a vacant lot at arrival.
Notes and issues	*This model uses arrivalTime, originCoordinates, prediction fallback and criteriaPriority as specified in the boundary/entity handover.*

UC-02 — Inspect Carpark Information

Field	**Specification**
Use case ID	UC-02
Use case name	Inspect Carpark Information
Owner	Nestor
Last updated	14 September 2026
Actors	Anonymous Driver
Description	The driver selects a recommendation and inspects the carpark information and evidence used to assess it.
Preconditions	1. UC-01 has produced at least one active recommendation.

Postconditions	1. The selected carpark information is displayed with the same values and evidence as the selected result. 2. The driver can request navigation, add the carpark to local favourites or open the anonymous report form. 3. No driver account or backend driver profile is created.
Priority	Must Have
Frequency	High - normally after a recommendation request
Main flow	1. The driver selects a carpark from the recommendation results. 2. The system opens the carpark detail view. 3. The system displays name, address, carpark type, payment modes, applicable rates, clearance information, current reported availability, predicted lots when available, confidence and the relevant source or retrieval timestamps. 4. The system displays the weighted score, ordered criteria, estimated cost, driving and walking measures, estimated-travel flag, likely-full indicator, explanation and warnings used for the result. 5. The system provides navigation, favourite and report actions.
Alternative flows	**UC-02.AC.1 - Selected result unavailable** 1. The selected result cannot be retrieved from the active search. 2. The system explains that the information is unavailable and returns the driver to the results. **UC-02.AC.2 - Evidence became stale** 1. A displayed value no longer satisfies its freshness rule. 2. The system labels it outdated or unavailable and advises the driver to refresh before relying on it. **UC-02.AC.3 - Driver requests navigation** 1. The driver selects Navigate and the system starts UC-03. **UC-02.AC.4 - Driver opens local action**

	1. The driver selects Favourite or Report. The system updates the local Favourite or opens the report form while retaining the detail view context.
Exceptions	**UC-02.EX.1 - Unexpected system error** 1. The system records the error without exposing sensitive details and keeps the driver in the detail or results view.
Includes	None.
Special requirements	● The detail view uses the same units, labels, timestamps, weighted score and evidence state as the UC-01 result. ● Availability and prediction are never represented as a guaranteed space. ● When data is already in the active result, the detail view should appear within one second for at least 95% of selections on a supported device.
Assumptions	● The selected carpark belongs to the active result set. ● A fallback from predicted to current availability remains visible with its reason. ● Navigation is provided by an external service.
Notes and issues	*None.*

UC-03 — Open External Navigation

Field	**Specification**
Use case ID	UC-03
Use case name	Open External Navigation
Owner	Nestor

Last updated	14 September 2026
Actors	Anonymous Driver; Device Operating System; Destination, Map and Routing Provider
Description	The driver requests directions to a selected carpark, and ParkChoiceSG hands the verified carpark destination to an external navigation service.
Preconditions	1. UC-02 is displaying a selected carpark with valid destination coordinates.
Postconditions	1. On success, an external navigation service opens with the selected carpark as destination. 2. On failure, the driver remains in ParkChoiceSG with the carpark details intact.
Priority	Must Have
Frequency	Medium - when a driver decides to travel to a result
Main flow	1. The driver selects Navigate. 2. The system obtains the verified carpark name and destination coordinates. 3. The system sends only that destination to a supported external navigation service through the device operating system. 4. The external service opens and calculates the route using its own origin and permissions. 5. ParkChoiceSG completes the handoff without providing turn-by-turn navigation.

Alternative flows	**UC-03.AC.1 - Multiple navigation services** 1. The operating system asks the driver to select a supported service. 2. The use case resumes at route launch. **UC-03.AC.2 - No navigation application** 1. The system keeps the detail view visible and presents the verified name, address and coordinates for manual use. **UC-03.AC.3 - External route failure** 1. The external service reports its own route failure; ParkChoiceSG keeps the selected details available.
Exceptions	**UC-03.EX.1 - Invalid destination** 1. The system does not launch navigation, explains the missing or invalid coordinates and keeps the detail view visible. **UC-03.EX.2 - Handoff error** 1. The system records a safe error message and keeps the detail view visible.
Includes	None.
Special requirements	● Only the selected carpark name and destination coordinates required for the handoff are sent. ● ParkChoiceSG does not need to collect a driver origin for the handoff; the external service controls origin and location permission. ● The external application launch should begin within two seconds for at least 95% of requests on a supported device.
Assumptions	● The device operating system supports external application links or intents. ● ParkChoiceSG does not control external routing accuracy, traffic, availability or permissions.

Notes and issues	*None.*

UC-04 — Submit Error Report

Field	**Specification**
Use case ID	UC-04
Use case name	Submit Error Report
Owner	Kayen
Last updated	14 September 2026
Actors	Anonymous Driver
Description	The driver submits an anonymous report about incorrect carpark information or a physical discrepancy.
Preconditions	1. The driver has a carpark detail context or a supported report entry point. 2. The installation can provide the quota key required by BR-6 without identifying the driver.
Postconditions	1. A permitted report is stored as an ErrorReport with category, description, submitted time and the originally supplied carpark identifier. 2. The separate ReportQuota count is incremented for the configured quota window. 3. No installation token or driver identity is stored on ErrorReport.
Priority	Should Have

Frequency	As needed when a discrepancy is observed
Main flow	1. The driver opens the report form. 2. The driver selects a category and enters a description; the system carries the selected carpark identifier when available. 3. ReportController validates required fields and checks ReportQuota using the installation token held separately from the report. 4. If allowed, the system stores the anonymous ErrorReport, increments the quota record and confirms submission. 5. The report becomes available to the authenticated operator workflow in UC-06.
Alternative flows	**UC-04.AC.1 - Quota reached** 1. The configured calendar-day or rolling-window quota is exhausted. 2. The system rejects the submission, explains the limit and does not store the token on a report. **UC-04.AC.2 - Missing report details** 1. The system identifies a missing category or description and asks the driver to correct it. **UC-04.AC.3 - Carpark identifier cannot be matched** 1. The system preserves the submitted identifier. UC-06 may automatically reject the report without inventing a Carpark association.
Exceptions	**UC-04.EX.1 - Submission failure** 1. The system does not claim success, displays a retry option and avoids exposing internal details.
Includes	None.
Special requirements	● The report contains no installation token, driver account, name or other identifying field. ● The quota key is maintained separately in ReportQuota and its window semantics must match BR-6.

	• The driver receives a clear confirmation or failure message.
Assumptions	• The device can securely provide a rotating installation token without user identity. • The report is based on an observed discrepancy and may require operator verification.
Notes and issues	*The configured quota window follows BR-6.*

UC-05 — Authenticate Operator

Field	**Specification**
Use case ID	UC-05
Use case name	Authenticate Operator
Owner	Kayen
Last updated	14 September 2026
Actors	System Operator / Administrator
Description	The operator supplies a username and password over TLS to unlock the administrative console.
Preconditions	1. The administrative endpoint is reachable over TLS 1.2 or later. 2. The operator has been provisioned an active OperatorAccount.
Postconditions	1. A signed session token is issued and stored in secure session storage. 2. The operator console becomes available until logout, expiry or revocation.

Priority	Must Have for operator workflows
Frequency	At the start of an administrative session
Main flow	1. The operator opens the login interface. 2. The operator enters the account username and password. 3. AuthenticationController validates the credentials against the server-side account store over TLS. 4. The system issues an OperatorSession token and displays the operator console.
Alternative flows	**UC-05.AC.1 - Invalid credentials** 1. The system rejects the attempt without identifying which field failed and keeps the login interface visible. **UC-05.AC.2 - Session expiry** 1. The system revokes the expired token and returns the operator to login.
Exceptions	**UC-05.EX.1 - Insecure or unavailable connection** 1. The system aborts authentication and displays a connection error without exposing credentials.
Includes	None.
Special requirements	● Authentication requests should complete within one second under normal service conditions. ● Password entry is masked and session tokens use secure storage.
Assumptions	● The operator has valid provisioned credentials. ● Each operator resolution and override references the responsible OperatorAccount.

Notes and issues	*Use username consistently in the boundary, control and entity models; do not use email as the attribute name.*

UC-06 — Manage User Error Reports

Field	**Specification**
Use case ID	UC-06
Use case name	Manage User Error Reports
Owner	Kayen
Last updated	14 September 2026
Actors	System Operator
Description	An authenticated operator reviews anonymous reports and resolves or rejects them with an audit record.
Preconditions	1. The operator has completed UC-05. 2. The report queue is reachable.
Postconditions	1. The report is marked Resolved or Rejected with mandatory remarks, resolution time and origin. 2. The reviewing OperatorAccount is recorded when the resolution is operator-originated. 3. An unmatched carpark identifier can be automatically rejected without a Carpark association.
Priority	Must Have for operator workflows
Frequency	On demand during an operator shift

Main flow	1. The operator opens the report queue. 2. ReportController retrieves pending reports in reverse chronological order. 3. The operator selects a report and reviews its category, description, timestamp, supplied identifier and any matched Carpark information. 4. The operator chooses Resolve or Reject and enters mandatory remarks. 5. The system records status, resolution origin, timestamp and reviewing account, then updates the queue.
Alternative flows	**UC-06.AC.1 - Carpark not in dataset** 1. The system cannot match the submitted identifier. 2. It rejects and archives the report with a system reason; no false Carpark association is created. **UC-06.AC.2 - Immediate correction required** 1. The operator starts UC-07 for the affected carpark while retaining the report context.
Exceptions	**UC-06.EX.1 - Empty remarks** 1. The system blocks the resolution and requests mandatory remarks. **UC-06.EX.2 - Queue unavailable** 1. The system reports the failure and leaves existing report state unchanged.
Includes	UC-05 - Authenticate Operator
Special requirements	● The queue can refresh without a full page reload. ● Resolved reports remain searchable for audit, subject to the agreed retention policy. ● Anonymous report data is not enriched with the quota token.
Assumptions	● Pending, Resolved and Rejected are the supported report statuses.

	• The model captures the final resolution rather than a complete moderation-history event log.
Notes and issues	*None.*

UC-07 — Apply Manual Carpark Override

Field	Specification
Use case ID	UC-07
Use case name	Apply Manual Carpark Override
Owner	Kayen
Last updated	14 September 2026
Actors	System Operator
Description	An authenticated operator applies or withdraws a closure or rate correction so erroneous source data does not reach recommendations.
Preconditions	1. The operator has completed UC-05. 2. The affected Carpark can be identified.
Postconditions	1. An OperatorOverride is recorded with author, reason and application time. 2. A closure excludes the carpark; a rate correction replaces the applicable RateSchedule rules. 3. The affected fields remain protected from source refresh until withdrawal.
Priority	Must Have for operator workflows

Frequency	When a discrepancy or physical carpark event occurs
Main flow	1. The operator selects a carpark by identifier or from a report. 2. The operator selects closure or rate correction. 3. For a rate correction, the operator enters replacement charging rules. The operator enters a mandatory reason for either override type. 4. The system records the override with the OperatorAccount and timestamp. 5. The system immediately applies the effective status to recommendation and detail queries and protects overridden fields from refresh.
Alternative flows	**UC-07.AC.1 - Withdraw override** 1. The operator selects an active override and chooses Withdraw. 2. The system records the withdrawing operator, time and reason and reconciles the affected values on the next refresh. **UC-07.AC.2 - Conflicting active override** 1. The system requires the earlier override to be withdrawn or superseded before another override of the same type becomes active.
Exceptions	**UC-07.EX.1 - Data-store write failure** 1. The system alerts the operator and preserves the previous effective state.
Includes	UC-05 - Authenticate Operator
Special requirements	● Overrides take effect across active recommendation queries after submission. ● A closure has no replacement rate rules; a rate correction has one or more rules with OperatorOverride provenance. ● Source refreshes preserve active overrides.
Assumptions	● The team adopts the proposed at-most-one-active-override-per-carpark-and-type rule.

	• Operator remarks and withdrawal data are retained for audit.
Notes and issues	*None.*

UC-08 — View API Health and Data Freshness

Field	**Specification**
Use case ID	UC-08
Use case name	View API Health and Data Freshness
Owner	Kayen
Last updated	14 September 2026
Actors	System Operator; Parking Data Provider
Description	An authenticated operator monitors health, freshness and cache evidence for each external data feed.
Preconditions	1. The operator has completed UC-05. 2. Health records are available to APIHealthMonitor.
Postconditions	1. Per-source health and freshness metrics are displayed without stopping background polling. 2. Stale or failing sources are visibly marked.
Priority	Should Have for operator workflows
Frequency	Continuous monitoring during operational shifts

Main flow	1. The operator opens the API health panel. 2. APIHealthMonitor reads SourceHealthRecord observations and SourceRefresh outcomes per DataSource. 3. The system displays request/failure counts, consecutive failures, last successful sync, cache age and the available provider or retrieval timestamps. 4. The system highlights sources that exceed configured freshness thresholds and provides a manual-refresh action that starts UC-09.
Alternative flows	**UC-08.AC.1 - Stale source** 1. The operator selects manual refresh and starts UC-09. **UC-08.AC.2 - Provider timestamp absent** 1. The system labels the recorded retrieval timestamp as retrieval time and does not call it a source update time.
Exceptions	**UC-08.EX.1 - Telemetry stream loss** 1. The console marks telemetry as unavailable while background services continue operating.
Includes	UC-05 - Authenticate Operator
Special requirements	● The panel refreshes automatically or through pull-to-refresh without a full-page reload. ● Unknown capacity, uptime or cache-age values remain absent rather than becoming zero.
Assumptions	● Each DataSource can have its own refresh frequency and freshness threshold. ● SourceHealthRecord and SourceRefresh are analysis entities; retention duration is decided later.
Notes and issues	*None.*

UC-09 — Trigger Manual Source Refresh

Field	Specification
Use case ID	UC-09
Use case name	Trigger Manual Source Refresh
Owner	Kayen
Last updated	14 September 2026
Actors	System Operator / Administrator; Parking Data Provider
Description	An authenticated operator requests an immediate refresh for a stale or failing external source.
Preconditions	1. The operator has completed UC-05 and is viewing the health panel. 2. The selected DataSource supports a manual refresh.
Postconditions	1. A SourceRefresh record captures the manual attempt and outcome. 2. On success, the cache and source health information are updated. 3. Active manual overrides remain effective.
Priority	Should Have for operator workflows
Frequency	On demand when a source breaches its freshness threshold

Main flow	1. The operator selects a stale source and requests refresh. 2. ExternalDataController records a manual SourceRefresh and sends the provider request. 3. The system validates the response, updates normalized availability/rate data and cache state, and records success metrics. 4. The system preserves active OperatorOverride records and updates the operator health panel.
Alternative flows	**UC-09.AC.1 - Refresh succeeds with partial data** 1. The system records the partial result, keeps unavailable fields absent and displays a warning. **UC-09.AC.2 - Refresh already running** 1. The system does not start a duplicate attempt and displays the existing refresh status.
Exceptions	**UC-09.EX.1 - External API failure** 1. The system records a failed SourceRefresh, retains the last trustworthy cache, increments failure metrics and does not advance the last-successful-sync time. **UC-09.EX.2 - Malformed payload** 1. The system discards the malformed payload, preserves the previous valid state and reports the failure to the operator.
Includes	UC-05 - Authenticate Operator
Special requirements	● Manual refresh may retry according to the provider policy but must respect external quotas. ● Active overrides are not overwritten by refresh data. ● Manual attempts reference the requesting OperatorAccount; scheduled attempts do not.
Assumptions	● The provider supplies usable HTTP status or equivalent health information.

	• Identifier reconciliation among availability and reference feeds remains an implementation risk.
Notes and issues	*None.*

Device-local requirement traceability

The current handover adopts device-local Option A. These capabilities use FR references until the SRS assigns additional use-case identifiers.

Capability	**Entity evidence**	**Boundary/control path**
Preferences and saved places	UserPreferences; SavedLocation	SettingsUI → PreferencesController
Recent searches	SearchHistoryEntry; maximum 10 entries	SearchHomeScreenUI → RecommendationController
Parking session and reminder	ParkingSession; at most one active session	SessionUI / NotificationInterface → SessionController
Favourite carparks	Favourite → Carpark	FavouritesUI → FavouritesController